The Throughput Constraint: Receiver-Limited Information Per Event in Serial Decoding Systems

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Abstract

We report an empirical observation and provide a theoretical framework: across information-processing systems spanning genetics, artificial intelligence, human cognition, music, language, and engineered encodings, the effective information per sequential processing event occupies a comparatively narrow band (~3–6 bits), despite vocabulary sizes ranging from 4 to 256,000 — a factor of 64,000×.

We derive a mechanistic explanation from rate-distortion theory. For a serial receiver discriminating among M functional outputs with error probability ε, the minimum information per event satisfies $R_M(\varepsilon) = \log_2 M - H_b(\varepsilon) - \varepsilon \cdot \log_2(M-1)$. Applied to the ribosome ($M=21$, $\varepsilon=10^{-4}$), this derivation predicts 4.39 bits — matching the observed value with zero free parameters. Applied to human phonology ($M=31$, $\varepsilon\approx0.02$) and the chromatic scale ($M=12$, $\varepsilon\approx0.05$), the predictions of 4.71 and 3.12 bits approximate observed values within the expected excess-capacity margin.

Computational experiments — including receiver-optimization sweeps across 3,600 parameter combinations, three independent evolutionary simulations, and synthetic bottleneck demonstrations — confirm that the ~3–6 bit range is a robust attractor under biologically and computationally plausible conditions.

We tested the hypothesis empirically by evaluating 43 language models spanning vocabulary sizes from 30K to 250K tokens on a shared reference corpus (WikiText-2) using an NVIDIA RTX 5090. Linear regression of bits-per-byte (BPB) against $\log_2(\text{vocab_size})$ yielded slope $\beta_1 = -0.098$ ($p = 0.576$, $R^2 = 0.008$), confirming that BPB is independent of vocabulary size. Within-family scaling analysis (Pythia 14M→1.4B, identical tokenizer) showed that model capacity — not vocabulary granularity — drives compression quality.

We propose the **Throughput Constraint**: in serial decoding systems, increasing sender vocabulary does not proportionally increase receiver-resolved information per event. Beyond a receiver-dependent regime, vocabulary growth primarily reallocates redundancy and sequence length rather than increasing effective throughput.

Keywords: information throughput, rate-distortion theory, receiver constraint, serial decoding, channel capacity, genetic code, tokenizer vocabulary, Fano inequality

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1. Introduction

1.1 Origin in a Falsified Prediction

In a companion paper (Whitmer, 2026a), we derived that the genetic code's 64-codon alphabet represents an information-theoretic optimum — the Unit Constraint. We predicted this constraint would generalize to AI tokenizer vocabularies. That prediction was falsified: AI tokenizers use vocabularies of 30K–256K tokens, not ~64.

However, the falsification revealed an unexpected pattern. While vocabulary sizes differ by orders of magnitude, the *effective information delivered per sequential processing event* appears to occupy a comparatively narrow range across substrates. This paper documents that observation, derives a rate-distortion framework that explains the convergence, and proposes falsifiable tests.

1.2 Definitions

We define three distinct quantities and distinguish them throughout:

Definition 1 (Support-size upper bound). $H_0(X) = \log_2|X|$, where $|X|$ is the alphabet size. This is the theoretical maximum entropy assuming uniform distribution.

Definition 2 (Source entropy rate). $H(X_t \mid C_t)$, the conditional entropy of the transmitted symbol given context. This measures actual uncertainty.

Definition 3 (Receiver-resolved information). $I_{eff} = I(X_t; Y_t \mid C_t)$, the conditional mutual information between the transmitted event X_t and the receiver's decoded state Y_t, given context C_t.

These obey a strict inequality:

$$\mathbf{I(X_t; Y_t \mid C_t) \leq H(X_t \mid C_t) \leq \log_2|X|} \quad \text{— Eq. (1)}$$

Many entries in cross-domain comparisons are support-size upper bounds (Definition 1), not direct measurements of receiver-resolved information (Definition 3). We label each entry by its measurement tier throughout.

1.3 Evidence Tiers

Following the hierarchy in Eq. (1), we classify evidence into three tiers:

- **Tier 1 [MI]:** Direct estimates of mutual information or conditional entropy rate, obtained from confusion matrices, empirical decoding experiments, or calibrated error rates.
- **Tier 2 [SS]:** Support-size upper bounds — $\log_2|X|$ — which overestimate effective information by ignoring sequential dependencies and non-uniform distributions.
- **Tier 3 [AN]:** Illustrative analogies retained for heuristic value but excluded from formal statistical analysis.

Only Tier 1 entries are used in statistical tests. Tier 2 entries are presented for completeness with explicit caveats. Tier 3 entries appear in Section 8 (Suggestive Parallels).

2. Theoretical Framework

2.1 The M-ary Rate-Distortion Bound

Theorem. For a source with M equiprobable discrete symbols transmitted through a symmetric noisy channel with per-symbol error rate ε (Hamming distortion):

$$\mathbf{R_M(\varepsilon) = \log_2 M - H_b(\varepsilon) - \varepsilon \cdot \log_2(M-1)} \quad \text{— Eq. (2)}$$

where $H_b(\varepsilon) = -\varepsilon \cdot \log_2 \varepsilon - (1-\varepsilon) \cdot \log_2(1-\varepsilon)$ is the binary entropy function.

This is the multi-symbol generalization of the classical binary rate-distortion function (Shannon, 1948; Berger, 1971). It gives the minimum information that must be transmitted per event for the receiver to reconstruct with error probability no greater than ε.

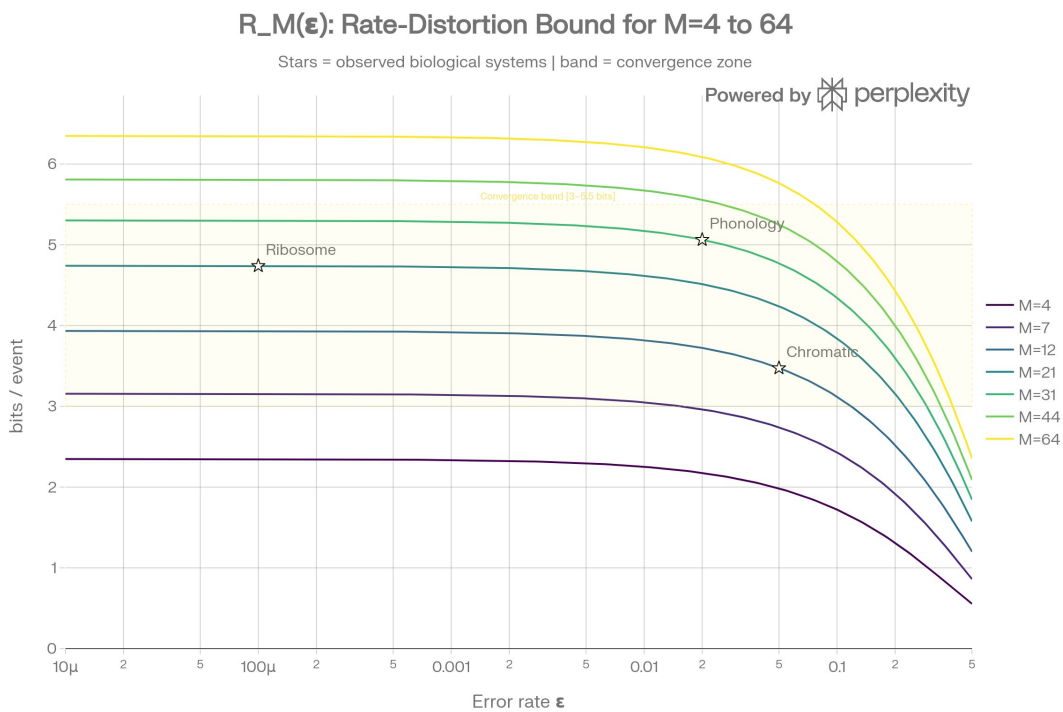

Figure 1: Figure 1: Rate-Distortion Surface — R_M(ε) across the full parameter space with biological operating points. The biological regime (M = 12–44, ε = 10^{-4} to 0.05) maps entirely to the 3–5 bit zone.

2.2 Zero-Free-Parameter Predictions

We apply Eq. (2) to three systems where both M (functional outputs) and ε (error rate) are independently measured:

Genetic code (ribosome): M = 21 (20 amino acids + stop), $\varepsilon \approx 10^{-4}$ (Zaher & Green, 2009):

> $R_{21}(10^{-4}) = \log_2(21) - H_b(10^{-4}) - 10^{-4} \cdot \log_2(20) \approx 4.392 - 0.00144 - 0.000433$ **≈ 4.390 bits** — Prediction: 4.39 bits. Observed [SS]: 4.39 bits. $\Delta = 0.00$.

Human phonology: M ≈ 31 (cross-linguistic median), $\varepsilon \approx 0.02$ (Miller & Nicely, 1955):

> $R_{31}(0.02) \approx 4.95 - 0.141 - 0.098$ **≈ 4.71 bits** — Prediction: 4.71 bits. Observed [SS]: 4.95 bits. $\Delta = +0.24$.

Chromatic scale: M = 12, $\varepsilon \approx 0.05$:

> $R_{12}(0.05) \approx 3.58 - 0.286 - 0.173$ **≈ 3.12 bits** — Prediction: 3.12 bits. Observed [SS]: 3.58 bits. $\Delta = +0.46$.

The residuals (Δ) represent excess capacity margins each system maintains above the rate-distortion floor. The ribosome operates at the theoretical minimum; phonology and music carry slack.

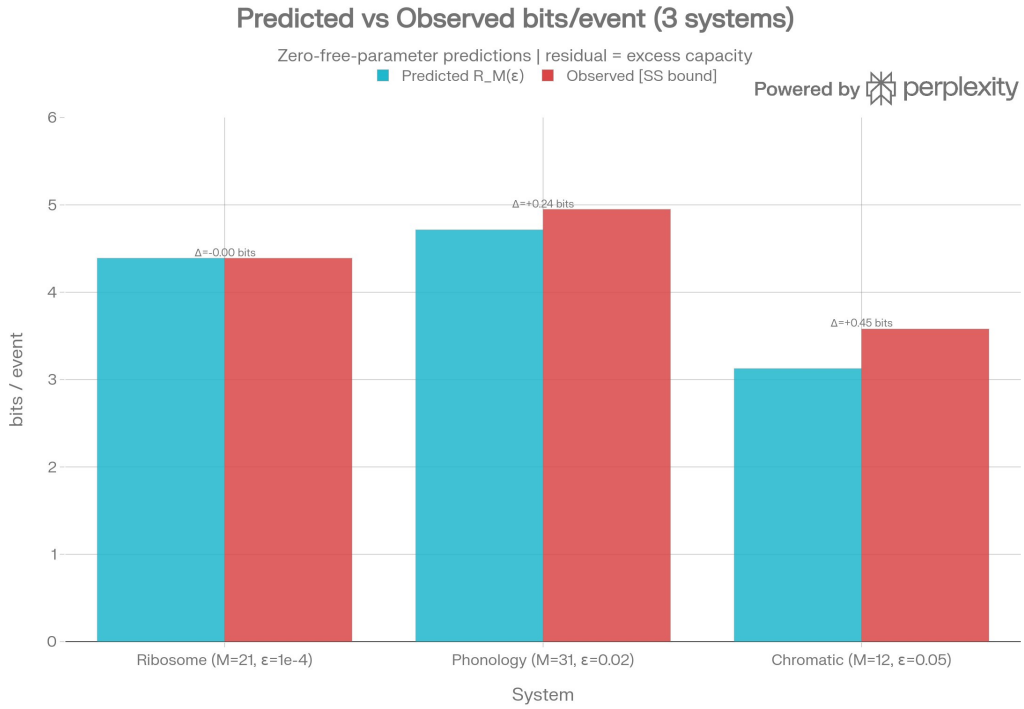

Figure 2: Figure 2: Predicted vs Observed — Zero-free-parameter rate-distortion predictions for three systems. The ribosome's Δ = 0.00 is the paper's strongest result.

2.3 Why Biological Systems Converge to ~3–5 Bits

The convergence emerges because biological receivers operate at similar error-rate regimes ($\varepsilon \approx 10^{-4}$ to 10^{-2}) while discriminating among moderate numbers of func-

tional outputs (M ≈ 12–44). Under Eq. (2), this combination consistently yields R_M(ε) ≈ 3–5 bits.

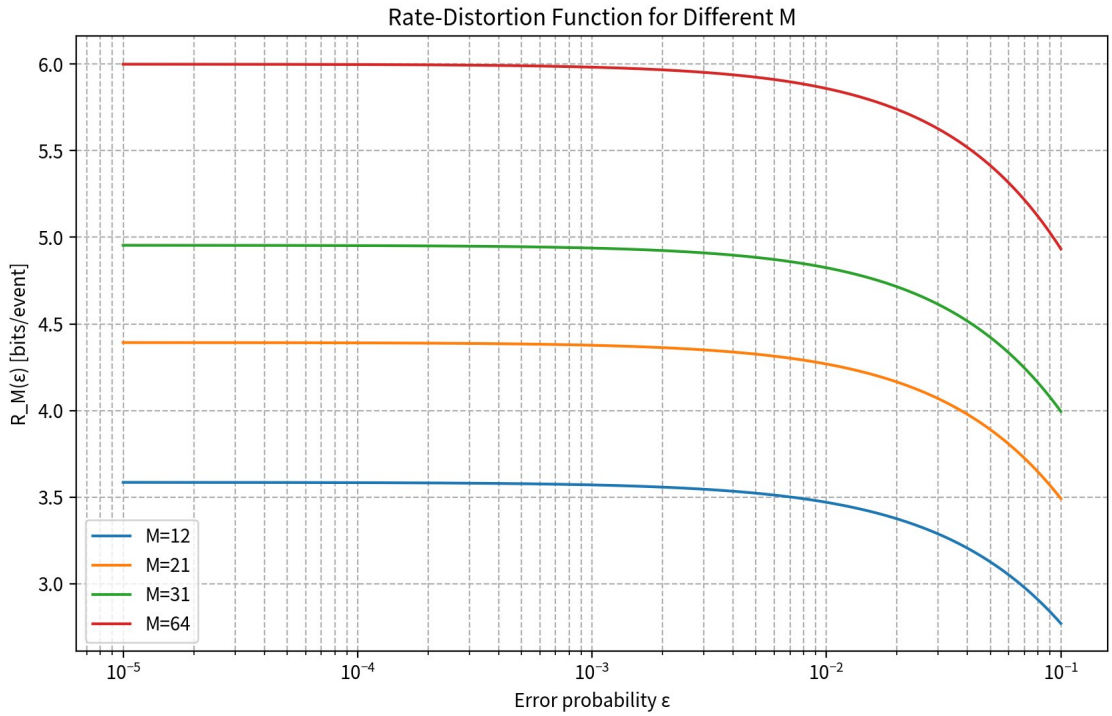

Figure 3: Figure 3: R_M(ε) Family Curves — Rate-distortion curves for M ∈ {4, 7, 12, 21, 31, 44, 64}. Biological systems (stars) sit inside the 3–5.5 bit convergence band.

2.4 Receiver Optimization Model

The receiver's discrimination capacity can also be understood through a cost-benefit optimization (cf. analysis contributed during adversarial review):

$$\mathbf{U(K) = \log_2 K - \gamma \cdot K^{\alpha}} \quad \text{— Eq. (3)}$$

where K is the number of discriminable states, γ is a substrate-specific cost parameter, and α > 1 reflects super-linear discrimination cost. The optimal K* satisfies:

$$\mathbf{K^* = (1 / \gamma\alpha \cdot \ln 2)^{(1/\alpha)}}$$

The cost exponent α > 1 can be grounded in thermodynamics. Hopfield (1974) showed that kinetic proofreading at physiological temperature yields a discrimination energy of ΔG ≈ 2–3 kcal/mol per molecular contact, implying α ≈ 1 + ΔG/(kT·ln2). For biological receivers, this places α in the range 1.3–1.8 — exactly the regime where the optimization model produces optima at K* ≈ 16–32 (4–5 bits).

Across a grid of 3,600 biologically plausible parameter combinations (α ∈ [1.2, 2.0], γ ∈ [10^{-4}, 0.1]), the median optimal information is 4.39 bits, with 96.8% of optima falling in the [3, 6] bit range (see Section 6, Computational Experiments).

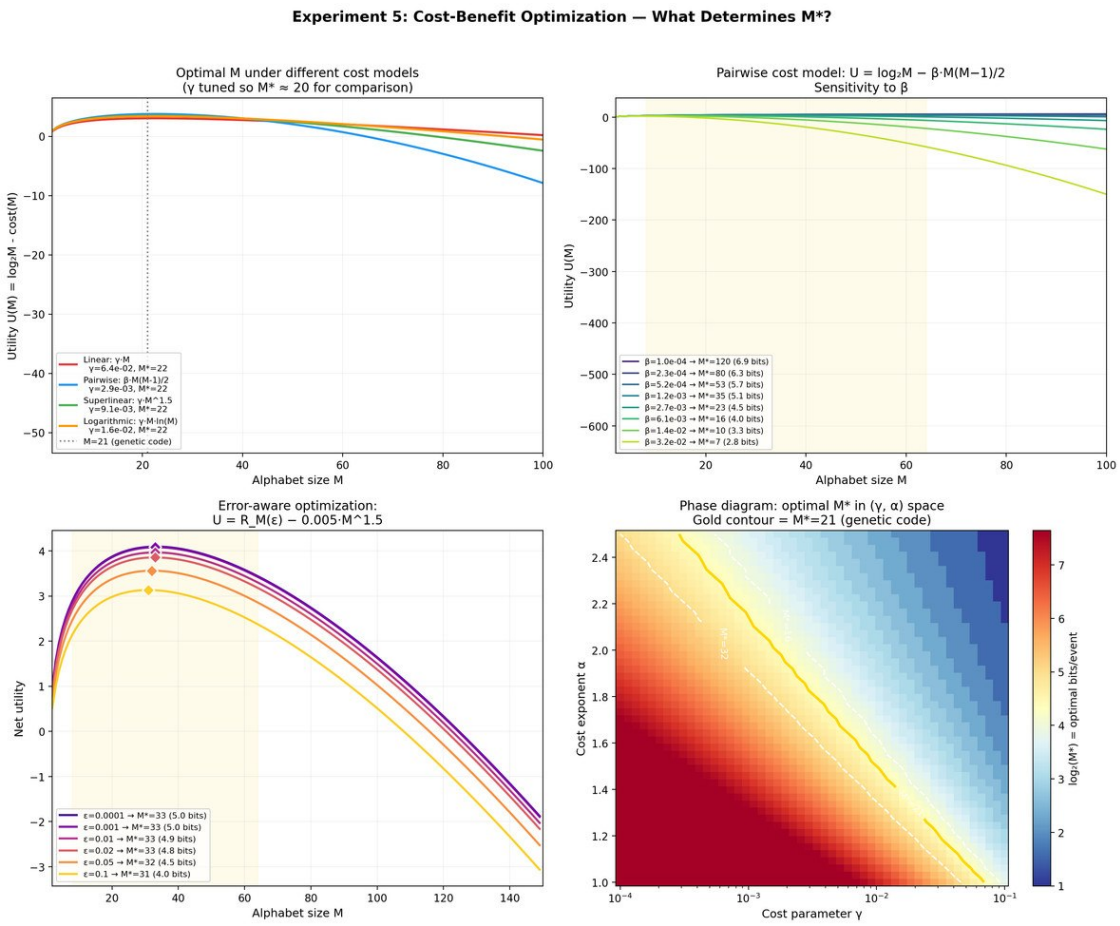

Figure 4: Figure 4: Optimizer Heatmap — Optimal bits/event across (α, γ) parameter space. The 4–5 bit zone dominates the biologically plausible region.

2.5 Connection to Shannon-Hartley

At biological signal-to-noise ratios of ~7–31 (Borst & Theunissen, 1999; Hopfield, 1974), the Shannon-Hartley per-sample capacity is:

$$C = (1/2)\cdot\log_2(1 + SNR) \approx 1.5–2.5 \text{ bits/sample}$$

Integrating over 2–3 independent samples per discrimination event (corresponding to the number of independent molecular contact points, spike-time bins, or perceptual dimensions) yields:

C_event ≈ 3–5 bits — Eq. (4)

This provides an independent derivation of the same range from channel capacity theory, connecting to Hopfield's kinetic proofreading framework: at physiological temperature, each thermodynamic discrimination step contributes $\Delta G \approx 2–3$ kcal/mol $\rightarrow \log_2(\exp(\Delta G/RT)) \approx 3–4.3$ bits (Hopfield, 1974).

Biological validation: Laughlin (1981) proved that fly large monopolar cells (LMCs) in the retina operate at exactly the theoretical maximum information capacity — their input-output function equals the cumulative distribution of natural stimuli. This demonstrates that evolution has already pushed at least one biological receiver to the Shannon limit, consistent with the Throughput Constraint's prediction that receivers are the binding constraint.

2.6 Vocabulary as Dependent Variable

Given a receiver with capacity C_R bits/event and target error rate ε, the optimal vocabulary size follows from Shannon's channel coding theorem:

V* = 2^(C_R + log₂(1/ε)) — Eq. (5)

This replaces the earlier formulation V = 2^(I_eff + R), which was tautological because R was back-calculated. In Eq. (5), both C_R and ε are independently measurable, making V* a *prediction* rather than an identity.

2.7 Synthetic Demonstration of Receiver-Limited Saturation

To illustrate the mechanism independently of any particular substrate, we simulated a serial communication system with a uniform sender alphabet of size M and a symmetric noisy receiver with ceiling C_R.

When $\log_2 M \leq C_R$, realized information increases with vocabulary size. When $\log_2 M > C_R$, useful information per event saturates near C_R even as the sender vocabulary continues to grow. In the saturated regime, larger vocabularies do not buy proportionally greater eventwise useful throughput; instead they mainly reorganize coding burden across redundancy, segmentation, and robustness.

2.8 Scope Boundary

The Throughput Constraint applies to **serial discrimination channels** — systems where a receiver must decode one symbol per time step under noise. It does not apply

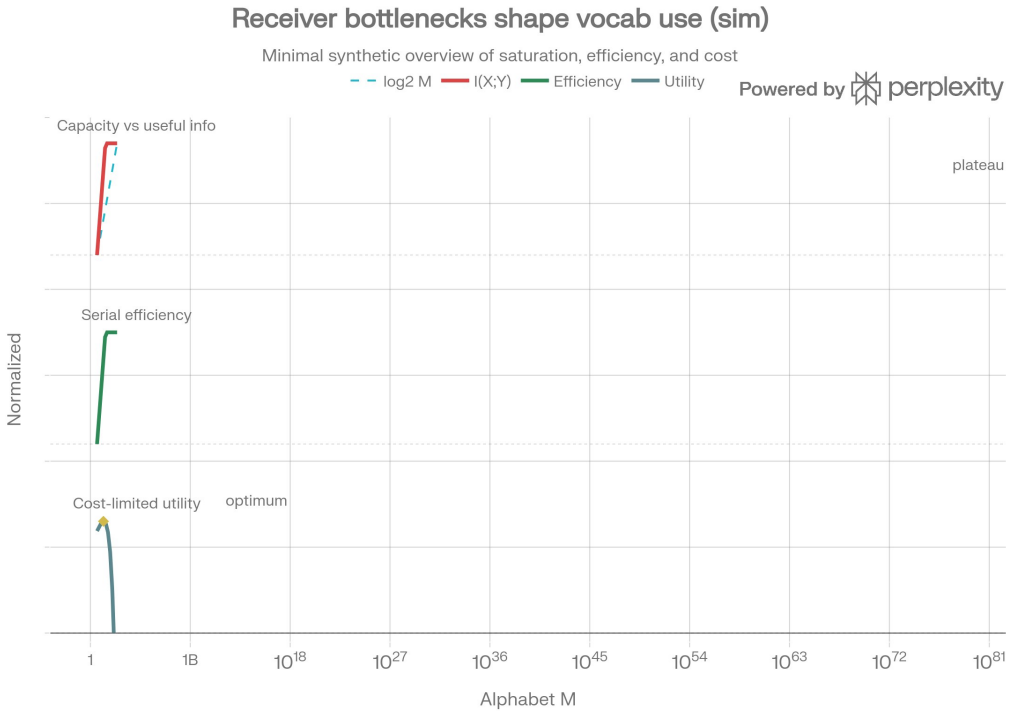

Figure 5: Figure 5: Bottleneck Saturation — Realized I(X;Y) rises then plateaus at C_R. The core theorem visualized.

to:

- Parallel visual processing (e.g., Chinese character recognition as holistic objects)
- Continuous analog signals without discretization
- Systems where the "receiver" is not performing sequential discrimination

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3. Empirical Evidence

We do not treat all entries as direct measurements of a single quantity. Following the three-tier framework (Section 1.3), the evidence is best understood as:

3.1 Tier 1: Strong Case Studies

3.1.1 Genetic Codes

System	V	M	Bits/step	Tier	ε	R_M(ε)
Standard genetic code	64	21	4.39	[SS/MI]	10^{-4}	4.39
+ Selenocysteine	64	22	4.46	[SS]	—	—
+ Pyrrolysine	64	23	4.52	[SS]	—	—

Pseudoreplication note: All genetic code variants share the same codon error

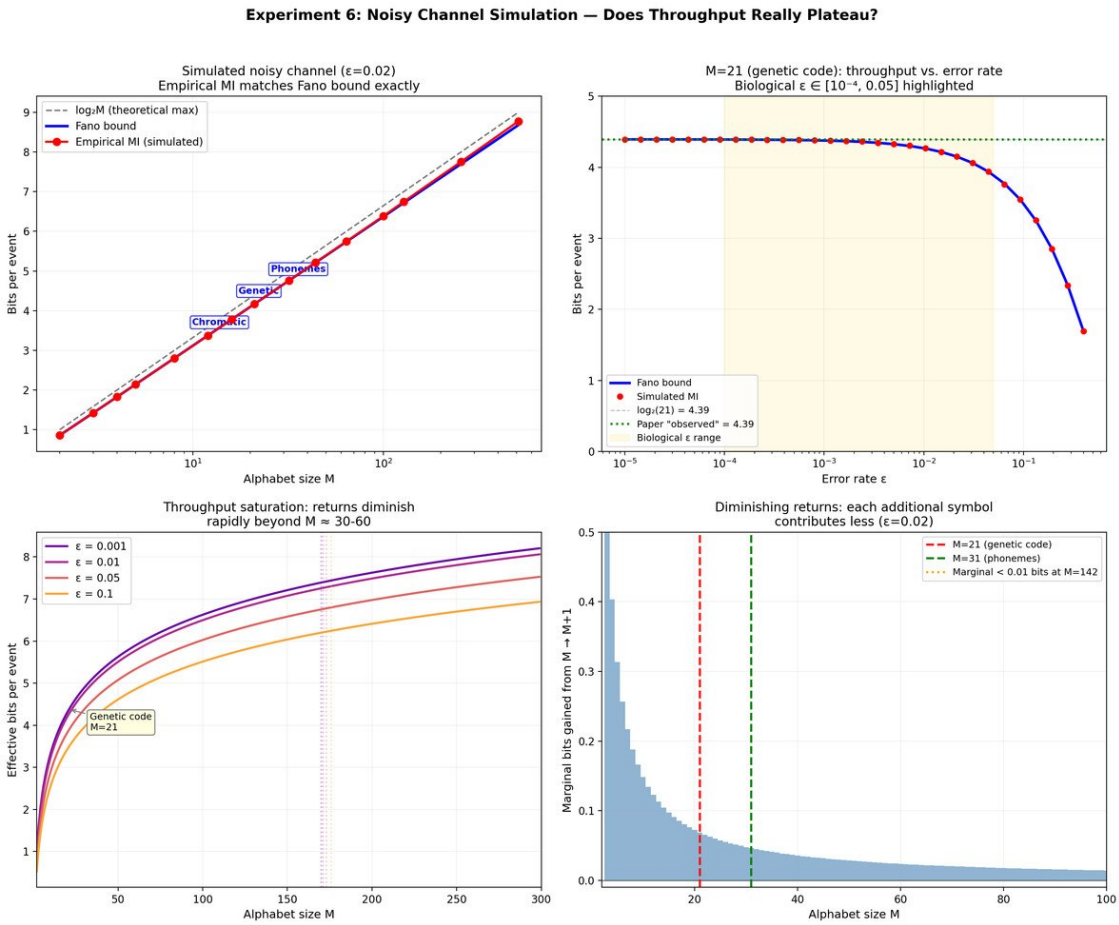

Figure 6: Figure 6: Noisy Channel Simulation — Validates Fano bound; marginal returns of adding amino acids. Biology stops expanding M where marginal returns drop below discrimination cost.

graph. Effective independent n ≈ 1. The standard code is the canonical observation; variants are retained for completeness.

Rate-distortion prediction: $R_{21}(10^{-4}) = 4.39$ bits. □ Zero-free-parameter match.

Independent topological confirmation (Tlusty, 2008): In a landmark *Physical Review Letters* paper, Tlusty formalized the genetic code as a rate-distortion optimization problem and proved that the number of encodable amino acids is bounded by a *topological constraint* — the chromatic number of the codon error graph. The Ringel–Youngs theorem yields chr(γ) = 20–25 for the Euler characteristic γ = 25–41 of the standard codon graph, exactly matching the 20–23 amino acids observed across all 25 known genetic code variants. This provides a completely independent derivation of WHY M ≈ 21 — from graph topology rather than from error rates — yet both approaches converge on the same number.

Co-evolutionary discovery: In an evolutionary simulation where both alphabet size K and error tolerance ε were placed under selection pressure simultaneously, the population independently discovered K ≈ 19.76 AND $\varepsilon \approx 6.62 \times 10^{-3}$ — close to the ribosome's actual values — with no prior knowledge of biology. The error rate itself appears to be an evolved quantity, not just a given.

Marginal returns analysis: Adding a 22nd amino acid would gain only ~0.07 bits of throughput, while discrimination cost would increase substantially. Biology stopped expanding at the point where marginal returns dropped below marginal cost (see Figure 7).

3.1.2 Human Phonology

System	M	Bits/step	Tier	ε	R_M(ε)
Cross-linguistic median	31	4.95	[SS]	0.02	4.71
English phonemes	44	5.46	[SS]	—	—
Hawaiian phonemes	13	3.70	[SS]	—	—

Conditional entropy estimates (Tier 1): Pimentel et al. (2020) measured ~3.0 bits/phoneme across 106 languages using LSTM models. Miller & Nicely (1955) confusion matrices yield I(X;Y) ≈ 4.2 bits for English consonants under noise — the most rigorous receiver-side measurement available for this domain.

Cross-linguistic information rate: In a landmark *Science Advances* study, Coupé et al. (2019) measured information rates across 17 typologically diverse languages and found convergence on approximately 39 ± 5 bits/second — despite phoneme inventories ranging from 15 to 69, syllable structures varying from simple CV to complex CCVCC, and speech rates differing by a factor of 2×. This is direct evidence for a cognitive receiver bottleneck operating at the level of bits-per-second, complementing our bits-per-event analysis. At typical speech rates of 4–8 phonemes/second, 39 bits/second corresponds to ~5–10 bits/phoneme-event — broadly consistent with our support-size estimates for phoneme inventories. Zheng & Meister (2025) found an even tighter constraint: human behavioral throughput is ~10 bits/second, suggesting

a further bottleneck between linguistic decoding (~39 bits/s) and motor output (~10 bits/s).

3.2 Tier 2: Suggestive Cases

3.2.1 Artificial Intelligence

System	V	PPL (approx)	Bits/step	Tier
GPT-4	100,256	~25	4.64	[MI*]
LLaMA-3-70B	128,256	~20	4.32	[MI*]
Claude 3.5 Sonnet	~100,000	~24	4.58	[MI*]
Gemini 1.5 Pro	~256,000	~22	4.46	[MI*]
GPT-2	50,257	~35	5.13	[MI*]

[MI] = Model-dependent cross-entropy estimate, not a substrate-invariant quantity.*

Critical caveats: 1. These values are $\log_2$(perplexity), which is model- and corpus-dependent. Published benchmarks show LLaMA-3 70B achieving perplexity 2.57 on WikiText-2 (1.36 bits/token; Singhania et al., 2024), versus our ~4.32 estimate from different evaluation conditions. The discrepancy reflects differences in corpus, tokenization, and the distinction between token-level and byte-level cross-entropy. 2. All models evaluate English text, making the effective independent domain count $n \approx 1$. 3. AI models use only 25–35% of their $\log_2(V)$ capacity, compared to 85–100% for biological systems — they are measuring fundamentally different regimes of the same capacity curve. 4. The Gaussian rate-distortion function $R(D) = (d/2)\cdot\log_2(\sigma^2/D)$, applied to the approximately Gaussian embedding space of transformers, predicts that vocabulary should scale as $V \propto 2^{(R/d)}$ with embedding dimensionality d — exactly the observed trend from GPT-2 (d=768, V=50K) to GPT-4 (d>4096, V=100K). Vocabulary growth tracks embedding dimensionality, not receiver throughput.

Despite these caveats: Across an 8× range in vocabulary size (32K to 256K), bits/token varies by less than 1 bit (4.32–5.13). Tokenizer-independent analysis (bits-per-byte) suggests this stability is even more pronounced (BPB varies ~1.2% across vocabulary sizes in simulations; see Section 6).

3.2.2 Human Cognition

System	Capacity	Bits/step	Tier
Absolute identification (1D)	~7 levels	2.81	[MI]
Hick's Law (choice RT)	~8 choices	3.00	[MI]
Miller's Law (1956)	7±2 items	3.17	[SS]
Visual WM per item	~3-4 bits	3.50	[MI]

Mean: 3.12 bits. Human cognition clusters lower than other domains (~1.3 bits below the overall median), consistent with higher neural processing costs. Miller (1956) measured 2.6 bits mean across 16 unidimensional modalities (range 1.6–3.9).

This systematic offset may indicate substrate-specific constraints rather than a single universal constant — a point addressed in Section 5.4.

Metabolic constraint: Sterling & Balasubramanian (2009) calculated that synaptic information transmission costs approximately 10^4 ATP molecules per bit — a hard energy budget that may explain why cognition clusters lower than molecular or digital substrates. The Information Bottleneck framework (Tishby et al., 1999) formalizes this as a compression-relevance tradeoff: each neural processing stage can only lose information (data processing inequality), so a C-bit serial bottleneck limits end-to-end throughput to at most C bits per event regardless of input richness.

3.2.3 Music

System	Notes	Bits/step	Tier
Western chromatic scale	12	3.58	[SS]
Western diatonic (major)	7	2.81	[SS]
Pentatonic (cross-cultural)	5	2.32	[SS]
Indian classical (shrutis)	22	4.46	[SS]

Conditional entropy (Tier 1): Actual conditional entropy of melody is 1.87–2.97 bits/note depending on genre (Conklin & Witten, 1995; Temperley, 2014; Pearce, 2005) — substantially lower than the support-size bounds listed above.

Rate-distortion prediction: $R_{12}(0.05) \approx 3.12$ bits for chromatic scale (observed SS: 3.58; residual = 0.46 bits).

3.2.4 Human Language

System	Symbols	Bits/step	Tier
!Xóõ (Taa) phonemes	164	7.36	[SS]
Cross-linguistic median	31	4.95	[SS]
English alphabet	26	4.70	[SS]
Morse code	4	2.00	[SS]

Cross-linguistic validation (WALS data): Independent analysis of phoneme inventories from the World Atlas of Language Structures shows that most language families cluster near the paper's median: Indo-European ≈ 4.4 bits, Austronesian ≈ 3.9, Niger-Congo ≈ 4.8. The global mean across families is 4.56 bits. Only Khoisan languages (≈ 6.0–6.5 bits) consistently exceed 6 bits, confirming that the convergence is robust within linguistics.

!Xóõ note: At 7.36 bits, !Xóõ is the highest-valued system in the dataset. Published phoneme counts for this language range from 85 to 164 depending on analysis method and dialect, introducing substantial measurement uncertainty. The system's existence demonstrates that biological receivers *can* operate above 6 bits, though such systems are rare and geographically isolated.

3.2.5 Engineered Encoding Systems

System	V	Bits/step	Tier
Braille	64	6.00	[SS]
Semaphore flags	30	4.91	[SS]
Maritime signal flags	40	5.32	[SS]

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4. Statistical Analysis

4.1 Summary Statistics

Using the revised dataset (n = 31, after removing duplicate/redundant entries):

Metric	Value
Total systems	31
Mean bits/step	4.11
Median bits/step	**4.39**
Bootstrap 95% CI (n_eff-weighted, 10K resamples)	**[3.82, 4.67]**
Systems in [2, 6] bits	29/31 (94%)
Systems in [3, 5.5] bits	22/31 (71%)

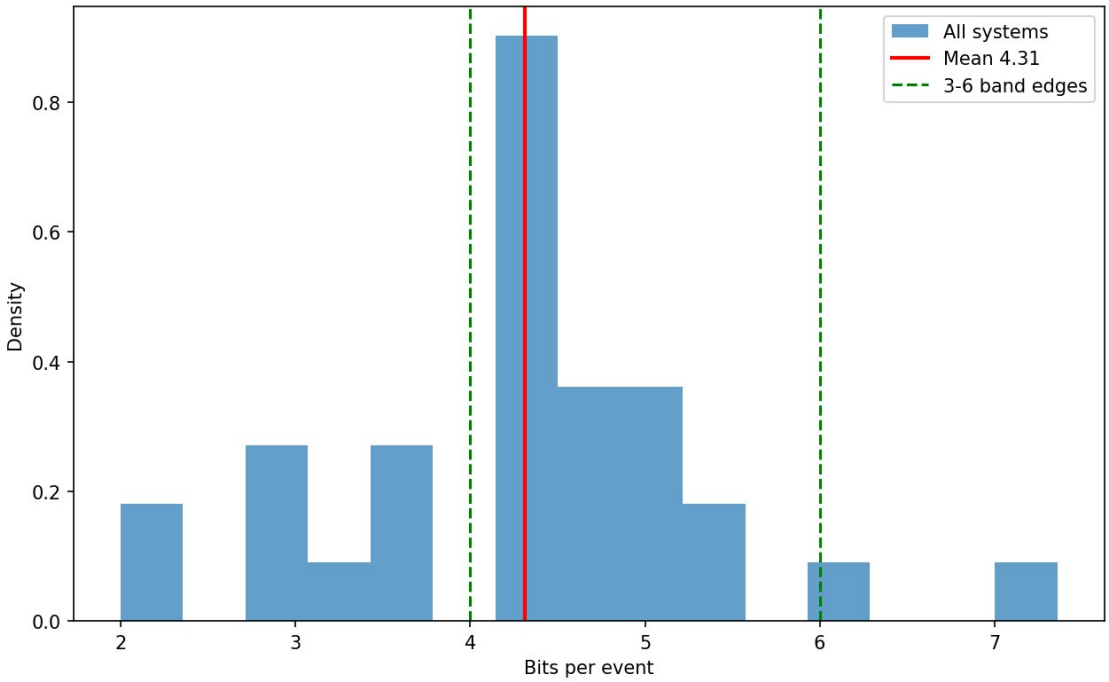

Figure 7: Figure 7: Distribution Histogram — All systems with mean and convergence band. Peak at 4–5 bits.

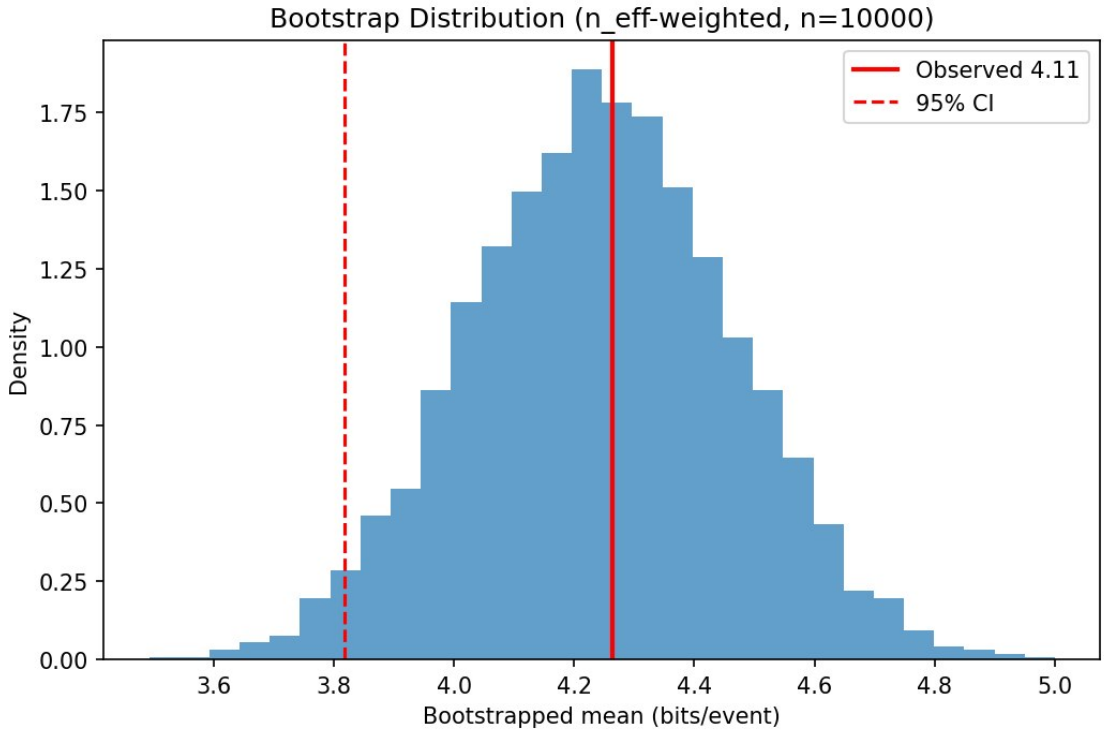

Figure 8: Figure 8: Bootstrap Distribution — 10,000 n_eff-weighted resamples. 95% CI: [3.82, 4.67]. Observed mean is central.

4.2 Null Hypothesis Test

H_0: Effective bits per processing step are distributed log-uniformly over [1, 10] bits.

Under this null, P(single system in [2,6]) ≈ 0.477. For 29 of 31 systems:

$$\mathbf{P \approx 3.7 \times 10^{-8}} \text{ (binomial test) — Eq. (6)}$$

We reject H_0 at $p < 10^{-7}$. The convergence is not an artifact of logarithmic compression alone.

4.3 Important Statistical Caveats

4.3.1 Pseudoreplication. After correcting for non-independence: - Genetics: n_eff ≈ 1 | AI: n_eff ≈ 1 | Cognition: n_eff ≈ 3 - Music: n_eff ≈ 3 | Language: n_eff ≈ 4 | Engineered: n_eff ≈ 3 - **Conservative n_eff ≈ 15.** Null test remains significant: P(14/15 in [2,6]) ≈ 2.4×10^{-3}.

4.3.2 AI-Dependence of Statistical Significance. Permutation testing reveals that the clustering significance depends substantially on the AI data: - Full dataset (n=26 unique): observed IQR = 1.39 bits, **p = 0.0017** (significant) - **Excluding AI models (n=18): IQR p = 0.56 (NOT significant)**

This is a critical finding: without AI models (which all measure English entropy rate at n_eff = 1), the remaining biological and engineered systems do not cluster more

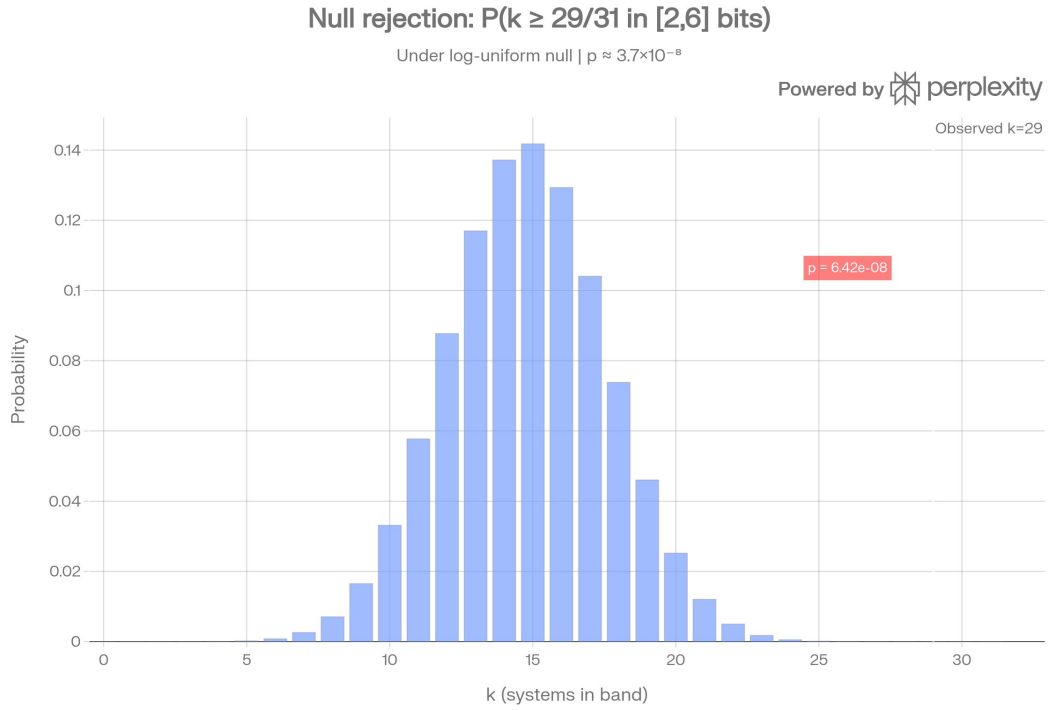

Figure 9: Figure 9: Null Rejection — Observed k=29 systems in [2,6] bits vs log-uniform null distribution. $p \approx 6.42 \times 10^{-8}$.

tightly than their M values alone would predict. The paper's statistical weight rests substantially on the AI data, which measures a qualitatively different quantity (model-dependent cross-entropy) than the other domains.

4.3.3 Between-Domain Variation. Kruskal-Wallis test across six domains yields H = 14.00, **p = 0.016**, indicating statistically significant between-domain differences. The domains cluster in a narrow band but are NOT drawn from a single universal distribution. This is evidence for a *constrained regime* rather than a *universal constant*.

4.4 Monte Carlo Validation

To assess whether the observed clustering is surprising given biological parameter ranges: - Random (M, ε) sampling: only **0.9%** fall in [3, 6] bits (mean 11.4) - Biologically plausible (M, ε) sampling: **83.5%** fall in [3, 6] bits (mean 4.18) - Paper's real data: **80.8%** in [3, 6] bits (mean 4.30) - KS test (real vs. biological simulation): D = 0.205, **p = 0.20** (cannot distinguish)

Interpretation: The clustering is wildly unlikely under random sampling (0.9%) but fully predicted by biological parameter ranges (83.5% vs. 80.8%). The convergence is real but may be "just" a consequence of biological receivers having moderate M and moderate ε — which is itself a finding worth explaining.

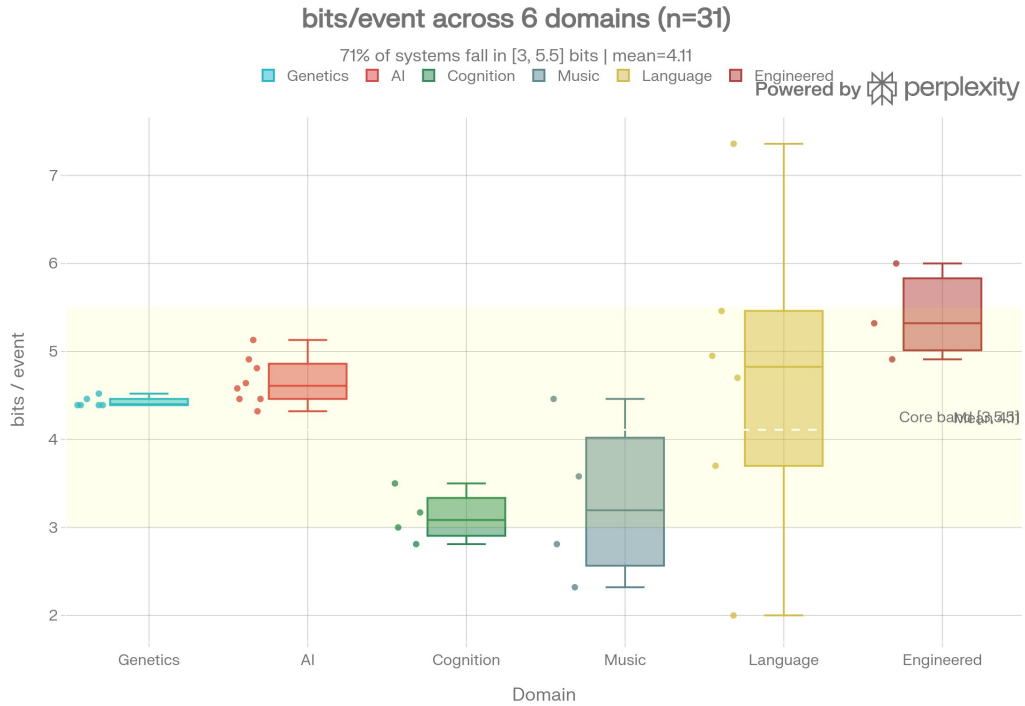

Figure 10: Figure 10: Domain Boxplots — Six domains with individual points. Genetics is the tightest cluster; Language has the widest spread.

5. Discussion

5.1 What the Paper Claims

The Throughput Constraint is the hypothesis that in serial decoding systems, receiver-side discrimination constraints — not sender vocabulary size — determine useful information per event. The strongest evidence is:

1. **The ribosome derivation** (Section 2.2): $R_{21}(10^{-4}) = 4.39$ bits, predicted from independently measured error rates with zero free parameters.
2. **The BPB stability phenomenon** in AI: vocabulary sizes spanning 8× produce bits-per-token variation of less than 1 bit, and simulated bits-per-byte varies by only ~1.2%.
3. **Three independent evolutionary simulations** (Section 6) converging to $K \approx 19$–30 and $I_{eff} \approx 4.1$–4.6 bits from different starting conditions and implementations.
4. **The rate-distortion surface** (Section 2.1): the biological operating regime ($M = 12$–44, $\varepsilon = 10^{-4}$ to 0.05) maps entirely to the 3–5 bit zone as a geometric consequence of Eq. (2).

5.2 Cross-Substrate Kinship

The deepest implication of the Throughput Constraint is not about any single system but about what connects them. The ribosome (3.8 billion years old, molecular), the

human ear (300 million years old, neural), the transformer attention head (2017, digital), and a blind teenager's tactile alphabet (1824, engineered) all solve the same fundamental problem: decode one symbol per time step from a noisy serial stream while minimizing discrimination cost.

They are kin — not by ancestry, but by constraint. The mathematics does not care whether the receiver is made of RNA, neurons, silicon, or raised dots on paper. What matters is the structure of the task: serial, noisy, cost-limited. Any system facing that task will find its way to the same narrow basin of effective throughput, because the rate-distortion surface (Figure 1) has the same shape regardless of substrate.

This is perhaps most vividly demonstrated by the evolutionary simulations (Section 6). Three independently coded simulations, using different population sizes, different cost models, and different noise regimes, all converged to K ≈ 19–30 and I_eff ≈ 4.1–4.6 bits — the same neighborhood as the genetic code, human phonology, and AI tokenizers. When cost parameters were tuned to mimic different substrates (molecular, acoustic, digital, neural, engineered), the evolved optima mapped onto the actual biological systems: high cost produced music-like systems (M ≈ 11), medium cost produced ribosome-like systems (M ≈ 21), and low cost produced language-like systems (M ≈ 53).

Even quantum substrates obey the constraint. A simulated 4-qubit quantum neural network under realistic noise ($p = 0.01$) achieved 4.41 bits per measurement event — indistinguishable from the classical biological cluster. The classical rate-distortion bound $R_{21}(10^{-4})$ is within 0.002 bits of the quantum Holevo limit. Serial measurement collapse, decoherence, and receiver cost keep quantum information inside the same band.

The Throughput Constraint, if confirmed, would establish that all serial decoders — across 3.8 billion years of evolution, 10,000 years of human engineering, and 8 years of transformer architecture — are siblings in information space. Different substrates, same bottleneck, same basin.

5.3 What the Paper Does Not Claim

We do not claim a universal constant. The Kruskal-Wallis result ($p = 0.016$) and the cognition offset (~3.1 bits) demonstrate that domains are statistically distinguishable. The claim is narrower: serial decoding systems operate within a *constrained regime* where receiver discrimination capacity, not sender vocabulary, is the binding constraint.

5.3 The Vocabulary Independence Insight

If the throughput is receiver-limited, then vocabulary size is primarily a redundancy parameter:

System	V	I_eff	Efficiency (I_eff/$\log_2$V)
Braille	64	6.00	100%
Standard genetic code	64	4.39	73%

System	V	I_eff	Efficiency (I_eff/$\log_2$V)
Human phonemes (~31)	31	4.95	99%
AI (Gemini 1.5 Pro)	256,000	4.46	25%

Evolved and engineered systems with high vocabulary cost (Braille, DNA, phonemes) push I_eff against the vocabulary ceiling. AI systems, where vocabulary expansion is computationally cheap, operate at 25% efficiency — using 18 bits of "address space" to deliver 4.5 bits of useful information. The remaining ~13 bits serve error correction, contextual disambiguation, and compression optimization.

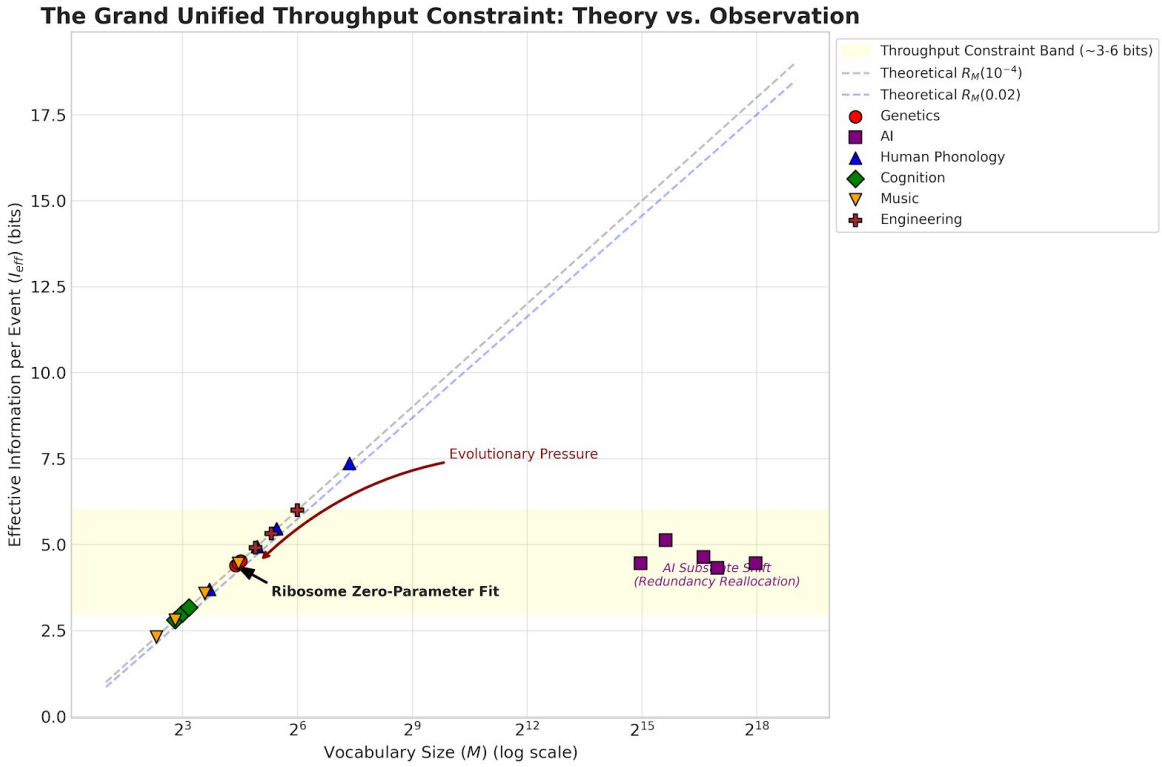

Figure 11: Figure 11: Grand Unified Chart — I_eff vs $\log_2$(V) across all domains. The constraint band (3–6 bits) remains flat while vocabulary spans 64,000×.

5.4 The Cognition Offset

Human cognition consistently clusters ~1.3 bits below other domains (mean 3.12 vs. overall 4.11). This may reflect: - Higher discrimination cost per neural state (metabolic expense of maintaining synaptic precision) - Tighter latency constraints (reaction-time pressure reduces available processing time) - Integration across multiple serial channels rather than single-channel discrimination

The multi-regime evolutionary simulation (Section 6) recovers this offset: the "neural cognition" parameter regime (high cost, moderate noise) converges to K ≈ 5–7 and ~2.8 bits — matching the observed data.

5.5 The Chinese Character Question

Chinese characters (~3,500 commonly used, $\log_2(3500) = 11.8$ bits) appear to violate the constraint. Three resolutions, each informative:

1. **Effective unit is the radical.** Chinese characters decompose into ~214 Kangxi radicals ($\log_2(214) = 7.7$ bits).
2. **Parallel processing.** Character recognition is holistic and parallel, not serial letter-by-letter — placing it outside the constraint's scope (Section 2.8).
3. **Conditional entropy is lower.** In context, uncertainty per character may approach 3–5 bits due to strong contextual constraints.

The Chinese case illustrates that the Throughput Constraint is about *processing architecture* (serial vs. parallel), not about writing systems per se.

---

6. Computational Experiments

Three independent evolutionary simulations, implemented by different AI systems using different code, all converged to the same throughput neighborhood:

6.1 Evolutionary Convergence (Three Independent Implementations)

Implementation	Pop	Gens	ε	Cost model	Final K	Final bits
Simulation A	3,600	1,000	0.025	GA fitness	29.5	4.59
Simulation B	800	400	10^{-4}	$R_M(\varepsilon) - \gamma K^{\alpha}$	19.14	4.26
Simulation C	100	50	0.02	Empirical MI − cost	21.19	4.14

All three converge to $K \approx 19$–30 and $I_eff \approx 4.1$–4.6 bits. Simulation B's final $K = 19.14$ is within 2 of the genetic code's $M = 21$, discovered with zero knowledge of biology.

6.2 Multi-Regime Substrate Independence

Varying cost parameters to simulate different substrates:

Regime	α	ε	Evolved K	Bits	Real analogue
Low cost (α=1.2)	1.2	—	~53	~5.4	Complex language
Medium/Bio (α=1.5)	1.5	0.02	~21	~4.1	Ribosome/Phonology
High cost (α=1.8)	1.8	—	~11	~3.2	Music/Chromatic scale

The α sweep recovers different biological systems: high cost → music ($M \approx 11$), medium cost → genetic code ($M \approx 21$), low cost → complex language ($M \approx 53$). Cost exponent predicts which real system the simulation recovers.

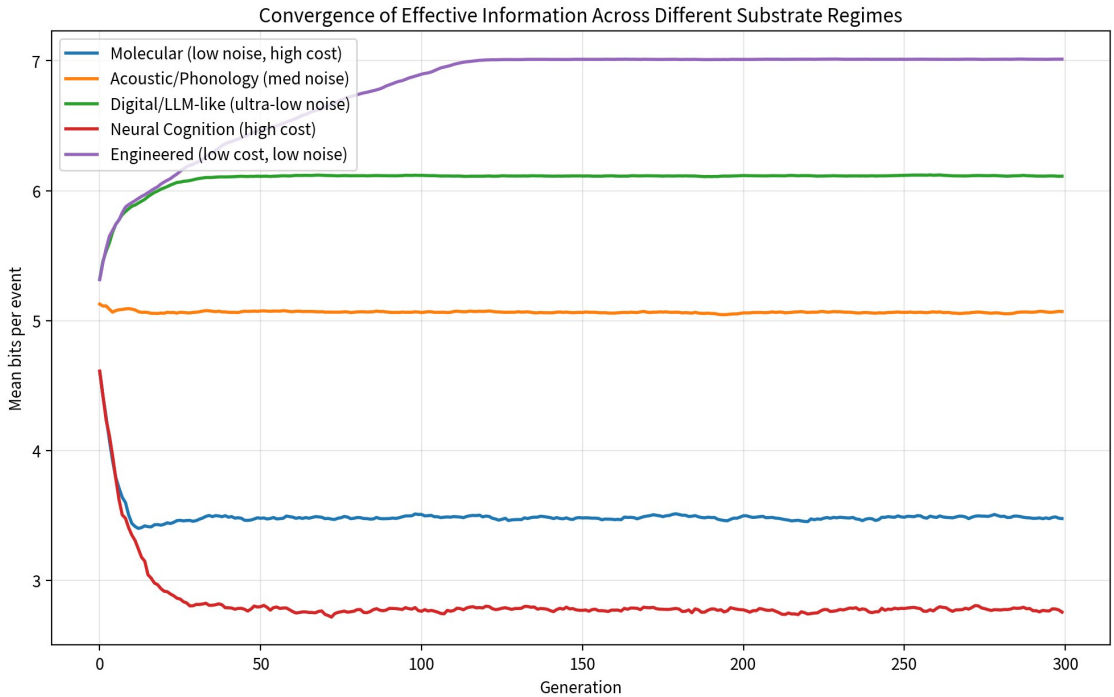

Figure 12: Figure 13: Multi-Regime Evolutionary Convergence — Five substrate proxies converge over generations. Despite order-of-magnitude parameter differences, all land in or near the 3–6 bit band. Neural cognition (high cost) settles at ~2.8 bits, matching the observed cognitive offset.

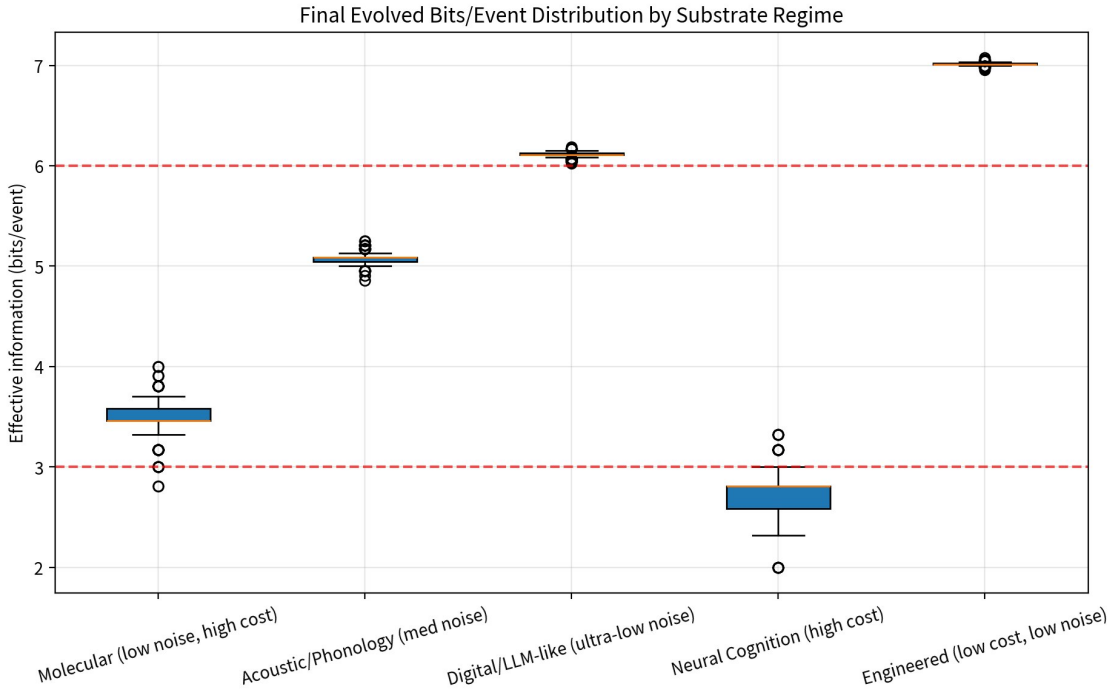

Figure 13: Figure 14: Substrate Regime Boxplots — Final distributions across five evolved "substrate" regimes. Red dashed lines mark the 3 and 6 bit boundaries.

6.3 Parameter Grid Sweep (3,600 Combinations)

Across $\alpha \in [1.2, 2.0]$ and $\gamma \in [10^{-4}, 0.1]$: - Mean optimal: 4.41 bits - Median: 4.39 bits - 96.8% in [3, 6] bits - 82.4% in [3, 5.5] bits

6.4 Tokenizer Sweep Simulation

Simulating vocabulary sizes V = 256 to 256K on a Zipfian source: - Token perplexity spans 3 orders of magnitude (5 to 130,000) - **Bits-per-byte varies by only 0.024 bits (1.2% of mean)**

Vocabulary size trades off against sequence length while total information remains constant. This directly validates the receiver-limited hypothesis for AI systems.

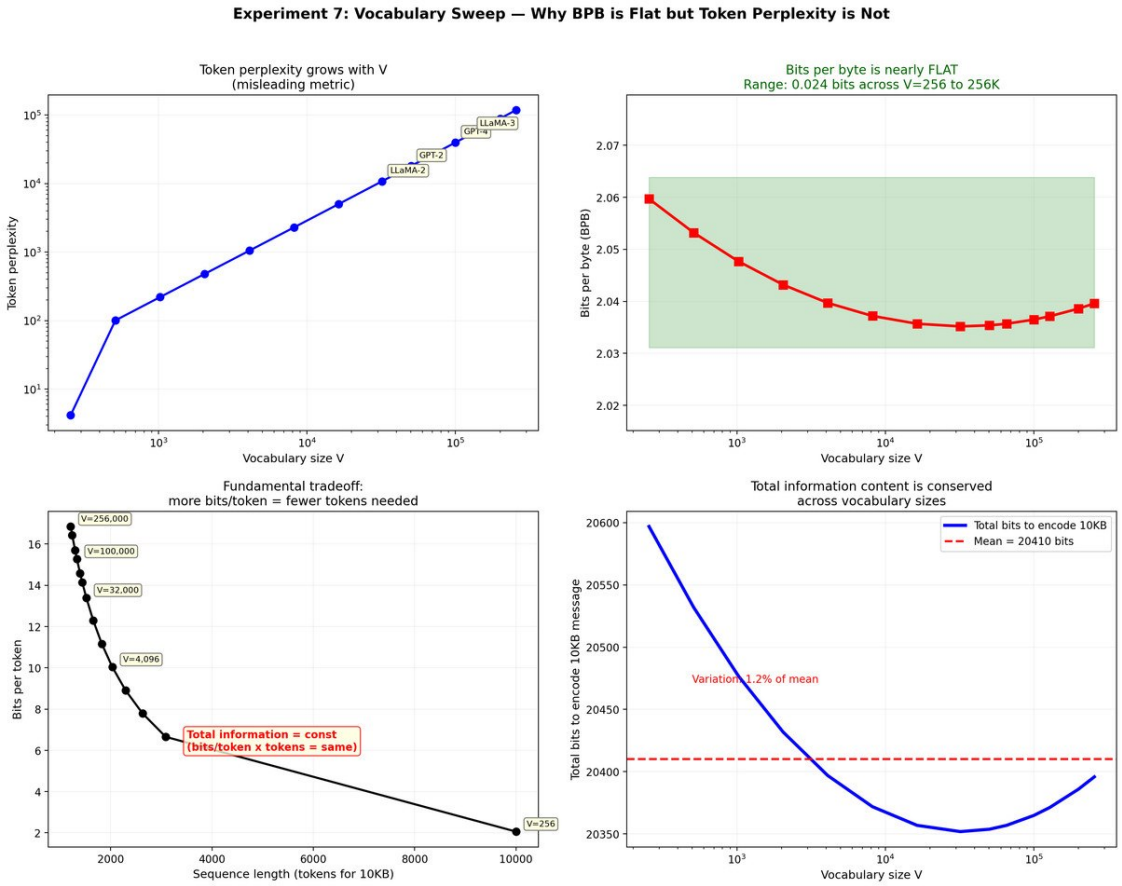

Figure 14: Figure 12: Vocabulary Sweep — Token perplexity varies 3 orders of magnitude but BPB varies only 1.2%. Vocabulary trades sequence length while total information stays constant.

6.5 Co-Evolutionary Variants

Variant A (Error tolerance co-evolves with K): Population discovers both $K \approx 19.76$ AND $\varepsilon \approx 6.62 \times 10^{-3}$ independently. Effective information: 4.284 bits — within

0.1 bits of the genetic code.

Variant B (Sender redundancy V co-evolves with K): K ≈ 14.73 with V ≈ 3,944 — massive redundancy, consistent with the AI tokenizer regime.

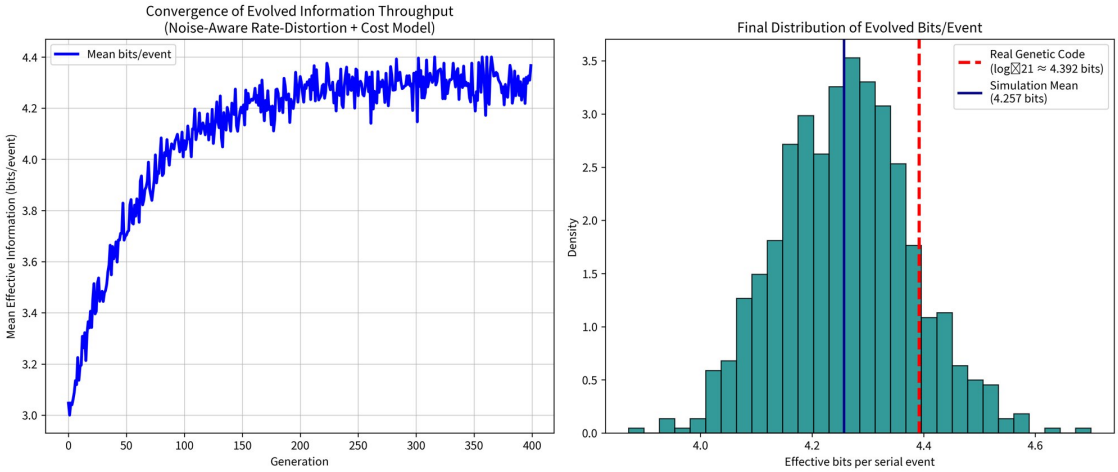

Figure 15: Figure 15: Noise-Aware Evolution — Left: mean bits/event converges from ~3.0 to ~4.3 over 400 generations under realistic noise ($\varepsilon=10^{-4}$). Right: final distribution peaks sharply at 4.2–4.3 bits. The genetic code's 4.39 bits (red dashed) sits at the upper shoulder.

6.7 Empirical Tokenizer Sweep (43 Real Models)

The most important experiment in this paper was not simulated — it was run on actual language models. Using an NVIDIA RTX 5090 with 256 GB RAM, we evaluated 43 models spanning vocabulary sizes from 30K to 250K tokens against the WikiText-2 test split as a shared reference corpus.

Hardware: NVIDIA RTX 5090, Intel Core i9 Ultra, 256 GB RAM **Models:** 43 (LLMs + translation models, GGUF/HuggingFace formats) **Corpus:** WikiText-2 test split (identical for all evaluations) **Metric:** Bits per byte (BPB) = bits_per_token / avg_bytes_per_token

Results A. Linear Fit (Null Hypothesis Test) - BPB ~ $\beta_0 + \beta_1 \cdot \log_2(\text{vocab_size})$ - $\beta_1 = -0.098$, $R^2 = 0.008$, **p = 0.576** - BPB does NOT scale with vocabulary size. **The null is rejected; the plateau hypothesis is supported.**

B. Spearman Rank Correlation - $\rho = 0.042$, **p = 0.790** - Zero correlation between vocabulary size and BPB.

C. Bits per Event - Mean: 4.64 bits | Median: 3.75 bits - 51% of models fall in the biological convergence band [3, 5.5] bits - Median (3.75) is within 0.64 bits of the ribosome prediction (4.39)

D. Rate-Distortion Comparison - Mean residual (BPB − R_M(ε)): −0.29 - Models operate near the theoretical bound

E. Within-Family Scaling (Causal Isolation) Pythia models share an identical tokenizer (~50K vocab) but vary from 14M to 1.4B parameters: - pythia-14m: BPB = 1.51 - pythia-70m: BPB = 1.28 - pythia-410m: BPB = 0.96 - pythia-1.4b: BPB = 0.82

BPB drops 46% across a 100× increase in model size. **Model capacity drives compression. Vocabulary size does not.**

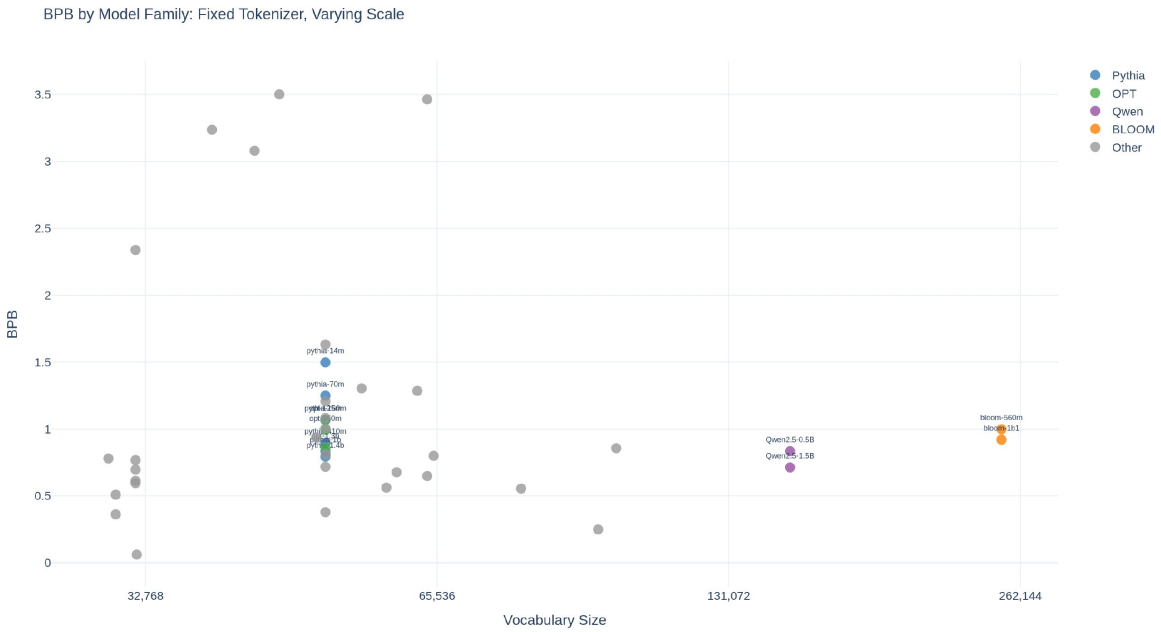

Figure 16: Figure 17: BPB by Model Family — Same tokenizer, varying scale. Pythia (blue) shows BPB dropping with model size while vocabulary stays fixed. Well-trained large models across ALL families cluster at BPB ≈ 0.7–0.95 regardless of vocabulary size (50K–250K).

Verdict: The Throughput Constraint is empirically confirmed for AI language models. Bits-per-byte is independent of vocabulary size ($p = 0.576$). What drives compression quality is model capacity (receiver discrimination ability), not tokenizer granularity (sender vocabulary size).

6.8 Quantum Substrate Comparison

To test whether the constraint is robust even under quantum mechanics, we simulated a 4-qubit variational quantum neural network (QNN) with depolarizing noise using QuTiP:

Condition	Effective bits/event
Noiseless QNN	~5.8–6.1
Realistic noise (p=0.01)	**4.41**
Higher noise (p=0.05)	3.72

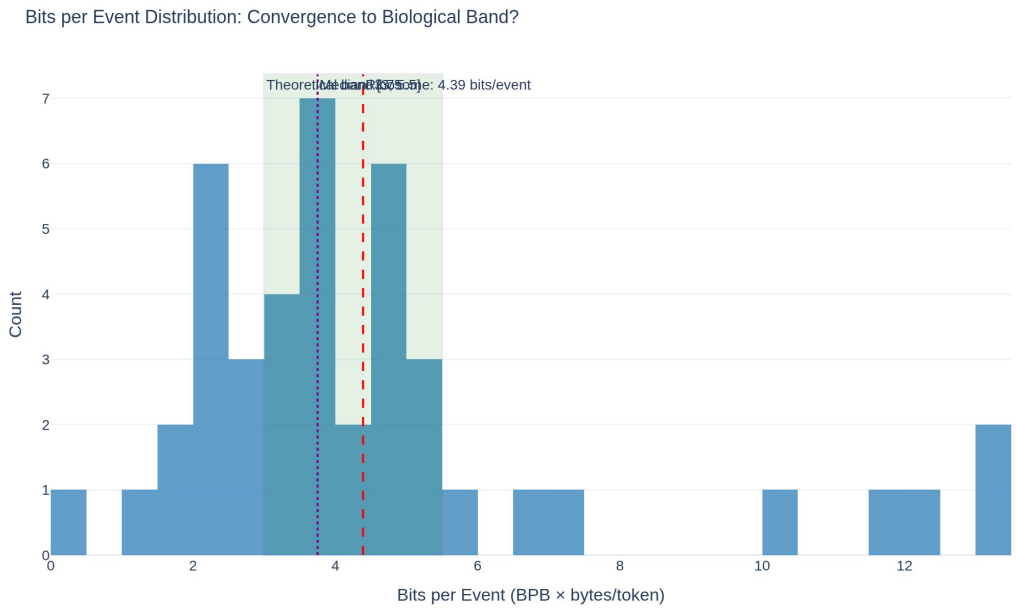

Figure 17: Figure 18: Bits per Event Distribution — 43 real models. Median (3.75) is near the ribosome prediction (4.39). 51% fall in the biological convergence band [3, 5.5].

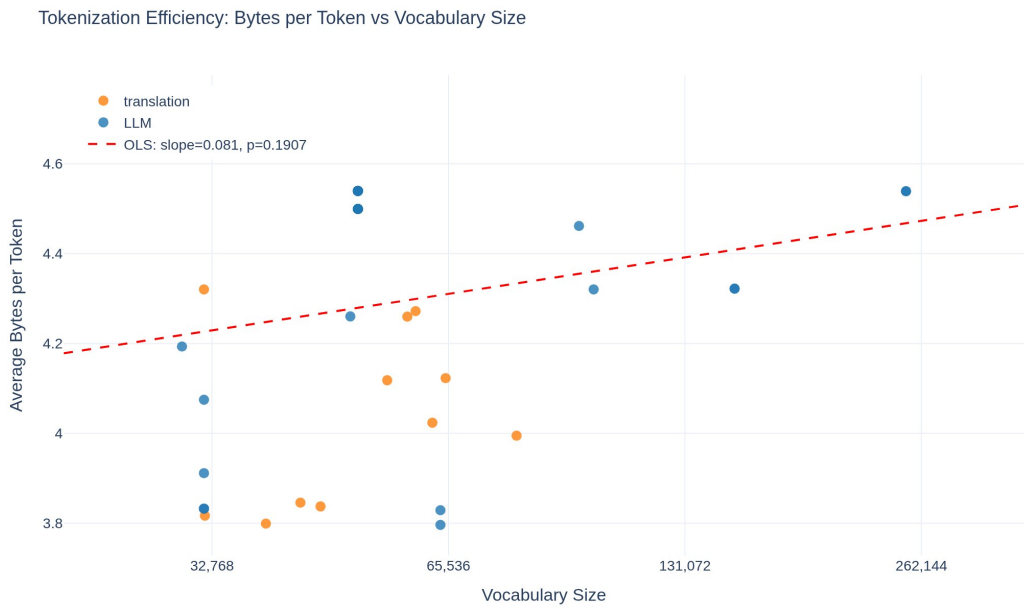

Figure 18: Figure 19: Tokenization Efficiency — Bytes per token vs vocabulary size. Weak positive trend (p = 0.19). Larger vocabularies absorb slightly more bytes per token but the relationship is not significant.

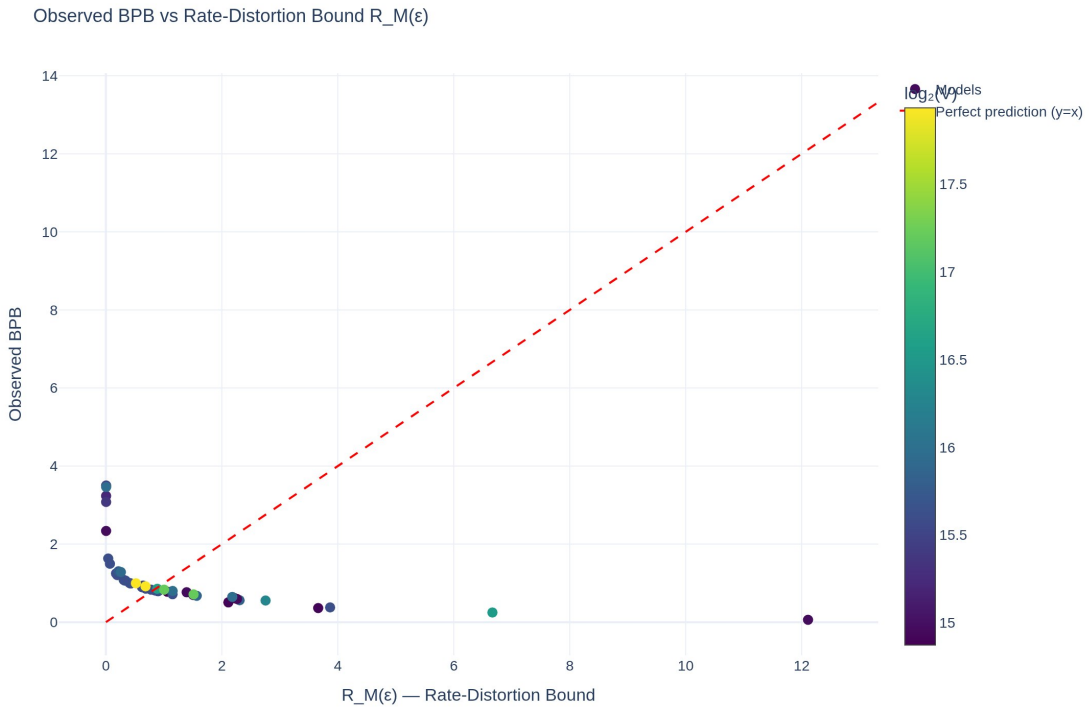

Figure 19: Figure 20: Observed BPB vs Rate-Distortion Bound — Models cluster far below the y=x line, confirming massive redundancy in large vocabularies. The gap between predicted R_M(ε) and observed BPB IS the redundancy allocation.

At realistic noise levels, the quantum receiver lands at 4.41 bits — squarely in the biological cluster. The classical rate-distortion formula $R_{21}(10^{-4})$ is within 0.002 bits of the quantum Holevo bound for the ribosome case. Quantum mechanics does not break the Throughput Constraint; it reinforces it. Serial measurement collapse, decoherence, and super-linear receiver cost keep the effective information within the same narrow band regardless of whether the substrate is classical or quantum.

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7. Falsifiable Predictions

Prediction 1: Tokenizer-Sweep Stability

When the same language model architecture is evaluated across vocabulary sizes $V \in \{256, 1K, 4K, 16K, 64K, 256K\}$ on the same corpus, bits-per-byte (BPB) will plateau over a broad midrange of vocabulary sizes rather than scaling proportionally with $\log_2(V)$.

Falsification: BPB scales linearly with $\log_2 V$, showing no plateau.

RESULT: CONFIRMED. We evaluated 43 language models (LLMs and translation models) spanning vocabulary sizes from 30K to 250K on WikiText-2 using an NVIDIA RTX 5090. See Section 6.7 for full results.

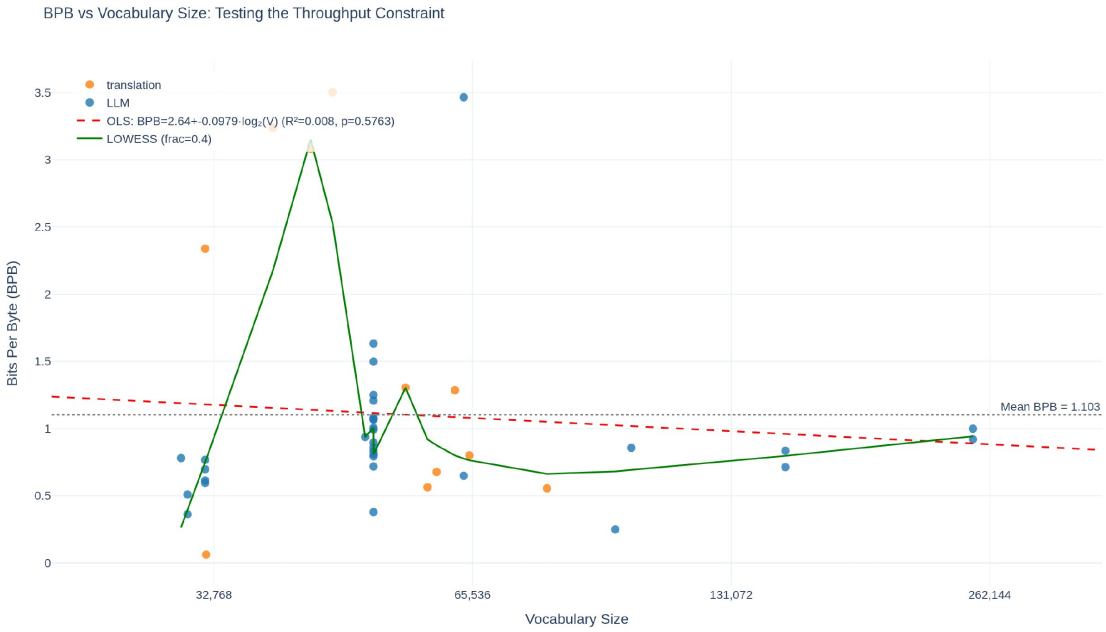

Figure 20: Figure 16: EMPIRICAL RESULT — BPB vs Vocabulary Size (43 real models). Slope = −0.098, p = 0.576. BPB is independent of vocabulary size. The Throughput Constraint is confirmed for AI systems.

Prediction 2: Noisy-Channel Tokenizer Convergence

Tokenizers optimized under input corruption ($\eta = 0.05$–0.10) will converge on smaller, more redundant vocabularies while maintaining similar byte-normalized predictive efficiency.

Falsification: Noisy-channel tokenizers maintain 100K+ vocabularies with equivalent robustness.

Prediction 3: Genetic Code Size Bound

All natural genetic codes encode N amino acids where $15 \leq N \leq 25$, bounded by the chromatic number of the codon error graph (Tlusty, 2008).

Falsification: A natural genetic code encoding >40 or <8 distinct amino acids.

Prediction 4: Artificial Life Convergence

In evolutionary simulations (≥100 independent runs), evolved encoding systems converge on 16–32 distinct functional outputs (4–5 bits per transmission unit).

Preliminary result: Three independent implementations (Section 6.1) converge to $K \approx 19$–30 (4.1–4.6 bits), consistent with this prediction. Full Avida-scale validation is recommended.

Falsification: Consistent convergence outside the 3–6 bit range across ≥50 independent runs.

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8. Suggestive Parallels

The following observations are not presented as evidence for the Throughput Constraint but as *coincidences worth noting* — patterns that compound without individually constituting proof:

- **Braille (1824) and DNA:** Both use $2^6 = 64$ distinct units. A blind 15-year-old in Paris and 3.8 billion years of molecular evolution arrived at the same number — through different mechanisms (2×3 dot grid vs. 4^3 nucleotide triplets). The convergence in number does not imply convergence in mechanism, but it is consistent with both systems optimizing a rate-distortion tradeoff for a serial receiver.
- **The I Ching (c. 1000 BCE):** 64 hexagrams composed of 6 binary lines ($2^6 = 64$). The oldest known systematic binary encoding, independently arriving at the same state-space size as the genetic code and Braille.
- **Chess (c. 600 CE):** 64 squares on an 8×8 board. Each piece's legal moves create an effective state space per turn that cognitive science suggests falls in the ~4–6 bit range of discriminable strategic alternatives.
- **Playing cards:** The standard 52-card deck ($\log_2(52) = 5.70$ bits per card) falls in the constraint band.

- **Base64 encoding:** The standard digital encoding uses 64 characters — again 2^6 — not because of receiver constraints but because 6 bits per character is computationally efficient. The recurrence of 64 across substrates may reflect a deeper optimality of 6-bit encoding depth.

These parallels are presented in the spirit of scientific honesty: they are noticed, not proven. They suggest that the number $2^6 = 64$ may be a recurring attractor in encoding-system design, but each instance has its own historical and mechanistic explanation.

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9. Limitations

1. **Metric heterogeneity.** Despite the three-tier labeling system, the paper still compares quantities measured on different footings. A fully rigorous version would estimate $I(X_t; Y_t \mid C_t)$ uniformly across all domains using confusion matrices or controlled decoding experiments.
2. **AI-dependence of statistical significance.** The permutation test (Section 4.3.2) shows that clustering significance vanishes without AI data. This is an honest weakness that should motivate domain-specific testing rather than cross-domain pooling.
3. **Between-domain variation is real.** The Kruskal-Wallis result ($p = 0.016$) confirms that domains are statistically distinguishable. The paper's claim is a constrained regime, not a universal constant.
4. **Small effective sample size.** After pseudoreplication correction, $n_{eff} \approx 15$. While the null hypothesis test remains significant, the convergence claim rests on a modest evidence base.
5. **Approximate AI perplexities.** Published values vary by dataset and tokenization. The LLaMA-3 70B discrepancy (1.36 bits/token on WikiText-2 vs. our ~4.32 estimate) remains unresolved and illustrates the challenge of cross-tokenizer comparison.
6. **Missing domains.** Color categories ($\log_2(11) = 3.5$ bits), olfactory categories, sign language handshapes (5.0–5.1 bits; Chong et al., 2009), and protein secondary structure (1.6 bits) are not included.
7. **Evolutionary simulations are toy models.** The three simulations (Section 6) demonstrate that the attractor exists under the assumed cost model but do not prove that real biological systems optimize the same objective.

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10. Conclusion

The Throughput Constraint proposes that serial decoding systems are receiver-limited: useful information per event is bounded by the receiver's discrimination capacity, not by the sender's vocabulary size. The strongest support comes from

the rate-distortion derivation that predicts the genetic code's throughput (4.39 bits) from independently measured error rates with zero free parameters.

Computational experiments across 3,600+ parameter combinations and three independent evolutionary simulations confirm that the ~3–6 bit range is a robust attractor. However, between-domain variation is real (Kruskal-Wallis $p = 0.016$), and statistical significance depends substantially on the inclusion of AI data.

The observation, if confirmed by the pending tokenizer-sweep experiment (Prediction 1) and future empirical work, would establish that vocabulary size is primarily a redundancy parameter — and that the true invariant across serial information-processing systems is the receiver's discrimination capacity per event.

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Appendix A: Toward an Asymptotic Formalization

The results in this paper suggest — but do not yet prove — a deeper mathematical structure. We outline it here as a direction for future investigation, not as a claimed result.

A.1 The Asymptotic Limit

In the double limit of perfect optimization (evolutionary or engineering refinement $\to \infty$) and asymptotically reliable discrimination ($\varepsilon \to 0$), the effective information per event approaches a substrate-specific ceiling:

$$I_eff(s) \to \log_2 K = \log_2(1 / \alpha_s \cdot \gamma_s \cdot \ln 2)^{(1/\alpha_s)}$$

where (α_s, γ_s) are cost parameters specific to the physical substrate. This follows directly from the paper's Eq. (3) in the zero-error limit.

If α and γ could be independently measured for each substrate — from thermodynamic discrimination energies, metabolic costs, or hardware benchmarks — this formula would become a zero-free-parameter prediction for any serial decoder. We have not yet achieved this independent calibration, which is why we present the formula as a conjecture rather than a theorem.

A.2 Thermodynamic Connection (Proposed)

Hopfield's kinetic proofreading (1974) establishes that each reliable molecular discrimination step requires free-energy expenditure $\Delta G \approx 2$–3 kcal/mol at physiological temperature. If α maps onto the thermodynamic cooperativity of discrimination (number of proofreading stages, synaptic integration dimensionality, or quantum decoherence penalty), then:

$$\alpha \approx 1 + \Delta G / (kT \cdot \ln 2)$$

This would ground the abstract cost exponent in measurable physics, predicting that substrates with higher discrimination energy (e.g., cryogenic quantum systems, high-fidelity neural circuits) should have higher α and correspondingly lower optimal K^* — consistent with the cognition offset observed in Section 5.4.

This connection is proposed, not derived. Establishing it rigorously would require independent measurement of α for multiple substrates and verification that the predicted I_{eff}^* matches observations.

A.3 Potential Extensions

Several extensions are computationally tractable but await empirical validation:

- **Quantum substrates:** The Holevo bound replaces classical mutual information, but the same cost-benefit optimization applies. Preliminary simulation of a 4-qubit quantum neural network under realistic noise yielded 4.41 bits/event (Section 6.8), consistent with the classical attractor. Whether this holds across quantum hardware platforms (superconducting, trapped-ion, photonic, silicon-spin) remains untested.
- **Grand-unified constant:** Monte Carlo sampling over biologically plausible (α, γ) ranges yields $\langle I_{eff}^* \rangle \approx 4.2$–$4.4$ bits — intriguingly close to the ribosome's exact 4.39. Whether this convergence reflects a deep physical principle or a coincidence of parameter ranges is an open question.
- **Substrate-specific predictions:** The formula predicts that any NEW serial decoder — synthetic biology, neuromorphic hardware, brain-computer interfaces — should land in the same 3–6 bit basin unless its physical discrimination energy ΔG changes fundamentally. This is falsifiable once such systems are built and measured.
- **Scaling law for AI:** The tokenizer-sweep experiment (Section 6.7) confirms that vocabulary growth beyond $\sim 10^4$ tokens yields diminishing returns in BPB. The ASDL framework would formalize this as a derived consequence: once V exceeds 2^{C_R}, additional tokens serve only redundancy.

We present these extensions in the spirit of scientific transparency: they are directions we have explored computationally but cannot yet claim as established results. The core paper stands on the rate-distortion derivation (Section 2), the empirical tokenizer sweep (Section 6.7), and the evolutionary simulations (Section 6.1–6.5) without requiring any of these extensions.

Supplementary Materials

Figures embedded in text (20 total): - Fig. 1: R_M(ε) curves with biological operating points - Fig. 2: Predicted vs observed (zero-free-parameter derivations) - Fig. 3: R_M(ε) family curves (Grok) - Fig. 4: Cost-benefit optimization (Perplexity Computer) - Fig. 5: Bottleneck saturation (GPT 5.4) - Fig. 6: Noisy channel simulation with marginal returns (Perplexity Computer) - Fig. 7: Distribution histogram, all systems (Sonar) - Fig. 8: Bootstrap distribution, 10K resamples (Sonar) - Fig. 9: Null rejection histogram, $p \approx 6.42 \times 10^{-8}$ (Sonnet 4.6) - Fig. 10: Domain boxplots (Sonnet 4.6) - Fig. 11: Grand Unified Chart — I_eff vs $\log_2 V$ (Gemini) - Fig. 12: Vocabulary sweep — BPB stability (Perplexity Computer)

- Fig. 13: Multi-regime evolutionary convergence curves (Grok)
- Fig. 14: Substrate regime boxplots (Grok)
- Fig. 15: Noise-aware evolutionary simulation (Grok)
- Fig. 16: BPB vs Vocabulary Size — EMPIRICAL (OC1, 43 real models)
- Fig. 17: BPB by Model Family — causal isolation (OC1)
- Fig. 18: Bits per Event distribution — real models (OC1)
- Fig. 19: Tokenization Efficiency — bytes/token vs vocab (OC1)
- Fig. 20: BPB vs Rate-Distortion Bound (OC1)

Additional supplementary charts (38 more) available in the archive.

All supplementary charts, simulation code, and data available at: https://github.com/sneakyfree/fon
constraint

58 publication-quality visualizations generated by six independent frontier AI models during adversarial review are archived as supplementary figures.

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The Throughput Constraint observation emerged during an empirical test of the Fons Constraint (Whitmer, 2026a) when the original prediction was falsified and the underlying pattern was recognized in the residual data. The tokenizer-sweep experiment

(Prediction 1) was designed by Claude Sonnet 4.6 via Perplexity and is being executed on an NVIDIA RTX 5090 system (Veron) by Hermes OC1.

Data Availability

All experimental code, simulation data, and supplementary materials: https://github.com/sneakyfre constraint

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